Data Log

Import

- dailyActivity_merged.csv- changed 'ActivityDate' to YYYY-MM-DD format
- Imported into schema bellabeat_case_study

Table	Description
dailyActivity_merged	Stats for a day. Activity, distance, calories
heartrate_seconds_merged	Timestamp with bmp
hourlyColories_merged	Calories burnt in an hour
hourlyIntenstities_merged	Activity intensity by hour.
hourlySteps_merged	Steps by hour.
minuteCaloriesNarrow_merged	Calories burnt in a minute.
minuteIntensitiesNarrow_merged	Activity intensity in a minute.
minuteMETsNarrow_merged	Metabolic Rate by Minute
minuteSleep_merged	Sleep by minute? Value 1 or 2.
minuteStepsNarrow_merged	Steps by minute.
weightLogInfo_merged	Weight, Fat, and BMI for users. One time entry.

- Data Fields and Types in data_types.csv
- Changed names. Replaces _merged with _to_april & _to_may
- Created backup tables in schema bellabeat_case_study_backup
- Deleted dailycalories_to_may, dailyintensities_to_may, dailysteps_to_may, minutecalorieswide_to_may, minuteintensitieswide_to_may, minutestepswide_to_may, and sleepday_to_may. All of these table formats were unique to May set of data and were redundant. Sleepday_to_may information was available in minutessleep_to_may. Wide data was available in narrow data and had same users. If any of this information is relevant, it makes more sense to combine the two sets of information, then restructure the timing.

Transform

There are 22 tables in the bellabeat_case_study schema. After merging the two time periods, there will be 11 if I delete the imported tables. Because there are no transformations being made to the data before the data is merged, and backups of the datasets have been created, I am okay with removing these tables to keep the database as clean as possible.

- Checking if users exist in both sets.

```
select distinct("Id")
from bellabeat_case_study.table
```

- May has 32 Ids and April has 34. Running this script will return values not shared by both sets.

```
select distinct "Id" from bellabeat_case_study.table_1
except
```

```sql
select distinct "Id" from bellabeat_case_study.table_2;

select distinct "Id" from bellabeat_case_study.table_2
except
select distinct "Id" from bellabeat_case_study.table_1;
```

Table Info	Unique Ids Between Sets
Daily Activity	2
Heartrate	2
Hourly Calories	3
Hourly Intensity	3
Hourly Steps	3
Minute Calories	3
Minute Intensities	3
METs	3
Sleep	3
Minute Steps	3
Weight Log	7

- Questions:
  - Is information by minute going to be useful for finding insights on general usage? *I can stick to hourly for now and dig deeper if necessary.*
  - Is there a significant risk of including ~3 users who are only present for half the timeframe? Are they likely to skew the data for a set with only ~35 people. *I can keep all Ids for now and filter inconsistent Ids if data appears muddled. With such a small sample, it's important to use as much of the information as possible.*
- Deleted minutecaloriesnarrow_to_april, minutecaloriesnarrow_to_may, minuteintensitiesnarrow_to_april, minuteintensitiesnarrow_to_may, minutestepsnarrow_to_april, and minutestepsnarrow_to_may from bellabeat_case_study schema
- bellabeat_case_study has 16 tables remaining.

```sql
-- tediously create new tables
create table bellabeat_case_study.activity_daily as
select *
from bellabeat_case_study.dailyactivity_to_april
union all
select *
from bellabeat_case_study.dailyactivity_to_may;

create table bellabeat_case_study.heartrate_seconds as
select *
from bellabeat_case_study.heartrate_seconds_to_april
union all
select *
from bellabeat_case_study.heartrate_seconds_to_may;

create table bellabeat_case_study.calories_hourly as
```

```sql
select *
from bellabeat_case_study.hourlycalories_to_april
union all
select *
from bellabeat_case_study.hourlycalories_to_may;

create table bellabeat_case_study.intensity_hourly as
select *
from bellabeat_case_study.hourlyintensities_to_april
union all
select *
from bellabeat_case_study.hourlyintensities_to_may;

create table bellabeat_case_study.steps_hourly as
select *
from bellabeat_case_study.hourlysteps_to_april
union all
select *
from bellabeat_case_study.hourlysteps_to_may;

create table bellabeat_case_study.mets_minute as
select *
from bellabeat_case_study.minutemetsnarrow_to_april
union all
select *
from bellabeat_case_study.minutemetsnarrow_to_may;

create table bellabeat_case_study.sleep_minute as
select *
from bellabeat_case_study.minutesleep_to_april
union all
select *
from bellabeat_case_study.minutesleep_to_may;

create table bellabeat_case_study.weightloginfo as
select *
from bellabeat_case_study.weightloginfo_to_april
union all
select *
from bellabeat_case_study.weightloginfo_to_may;
```

- Delete old tables from bellabeat_case_study for useability. Data is still available in bellabeat_case_study_backup

Validation

- Check heartrate_seconds and mets_minute because they are too large for Excel. If count is less than total rows, null values exist.

```sql
select count(*) as "Total Rows",
count("Id") as "Id",
count("Time") as "Time",
count("Value") as "Value"
```

```
from bellabeat_case_study.heartrate_seconds;

select count(*) as "Total Rows",
count ("Id") as "Id",
count("ActivityMinute") as "Activity Minute",
count("METs") as "METs"
from bellabeat_case_study.mets_minute;
```

Output:

Total Rows	Id	Time	Value
3638339	3638339	3638339	3638339

Total Rows	Id	Activity Minute	METs
2770620	2770620	2770620	2770620

- weightloginfo has 97/101 NULL values for Fat. *Not logging fat*.

Analysis

- Sleep- FitBit api collects sleep data where 1= asleep, 2 = restless, 3= fully awake. Only pulls data when sleep tracking is active.

```
select extract(hour from date) as hour_of_day, avg(value) as avg_sleep_value,
count(*) as value_count
from bellabeat_case_study.sleep_minute
group by hour_of_day
order by hour_of_day;

select value, count(*)
from bellabeat_case_study.sleep_minute
group by value;
```

Output:

hour_of_day	avg_sleep_value	value_count
0	1.07443602422681920590	34177
1	1.0711169964168874	38514
2	1.07458458315478741132	42006
3	1.06200468203002512833	46561
4	1.06916458880876968349	44517
5	1.06925490768444866088	41109
6	1.10185449358059914408	31545
7	1.11785937725588334376	20779
8	1.12793897931902178187	12717
9	1.0894911787266684	7822
10	1.1146907216494845	4656
11	1.1680327868852459	2196
12	1.1882352941176471	1105
13	1.1508474576271186	1180
14	1.1157989228007181	1114
15	1.1798344620015049	1329
16	1.1608730800323363	1237

17	1.1547619047619048	840
18	1.3379204892966361	654
19	1.2521315468940317	821
20	1.3421446384039900	2005
21	1.2194520945064502	6899
22	1.16012140733399405352	16144
23	1.09936286966449379442	27153

value	count
1	356006
2	27033
3	4041

- Looking at Metabolic Equivalent of Task to see activity patterns. This data is continuously logged.

```
select extract(hour from "ActivityMinute") as hour_of_day, avg("METs") as
avg_METs, count(*)
from bellabeat_case_study.mets_minute
group by hour_of_day
order by hour_of_day;
```

Output:

hour_of_day	avg_mets	count
0	10.8973913043478261	117300
1	10.5378113271920846	117240
2	10.3483602290473712	115260
3	10.1938706968933669	119100
4	10.2365470085470085	117000
5	11.6529844364631435	116940
6	12.4985362095531587	116820
7	14.0211739502999143	116700
8	15.2107191898386543	116520
9	16.1953889747552808	116460
10	16.5458376511226252	115800
11	16.5050762300762301	115440
12	17.3452989224887035	115080
13	16.8658224543080940	114900
14	17.0556687565308255	114840
15	15.9763139514580059	114540
16	16.8994830003505082	114120
17	17.7185087719298246	114000
18	17.9972792697911181	113940
19	17.8846322625943479	113940
20	15.6512827271129854	113820
21	14.2548083684950774	113760
22	13.1090989088349173	113640
23	11.8018773135906928	113460

- Seeing how activity time is spread by users and overall.

```sql
select round(avg("SedentaryMinutes")/60, 2) as avg_sed_hrs,
round(avg("LightlyActiveMinutes")/60, 2) as avg_light_hrs,
round(avg("FairlyActiveMinutes")/60,2) as avg_moderate_hrs,
round(avg("VeryActiveMinutes")/60, 2) as avg_very_active_hrs
from bellabeat_case_study.activity_daily
group by "Id";
```

```sql
select round(avg("SedentaryMinutes")/60, 2) as avg_sed_hrs,
round(avg("LightlyActiveMinutes")/60, 2) as avg_light_hrs,
round(avg("FairlyActiveMinutes")/60,2) as avg_moderate_hrs,
round(avg("VeryActiveMinutes")/60, 2) as avg_very_active_hrs
from bellabeat_case_study.activity_daily;
```

Output:

avg_sed_hrs	avg_light_hrs	avg_moderate_hrs	avg_very_active_hrs
21.10	2.30	0.30	0.13
18.33	2.81	1.38	0.00
18.19	4.59	0.18	0.14
20.47	2.97	0.04	0.02
21.09	2.35	0.06	0.09
12.47	3.91	0.23	0.07
12.18	2.51	0.49	1.43
14.00	3.44	0.02	0.10
11.35	3.87	0.00	0.00
20.26	0.98	0.02	0.02
19.31	2.12	0.02	0.01
10.86	3.33	0.23	0.39
21.03	0.57	0.01	0.09
20.45	1.03	0.39	0.22
13.57	4.86	0.04	0.02
11.43	4.22	0.36	0.21
18.99	2.53	0.17	1.42
15.82	3.04	0.27	0.31
13.89	3.72	0.30	0.63
16.86	1.73	0.07	0.02
22.37	0.83	0.08	0.02
19.32	1.73	0.11	0.08
21.88	1.58	0.18	0.25
14.35	2.20	0.39	0.63
11.79	2.66	0.17	0.96
18.00	5.24	0.07	0.17
11.80	3.07	0.88	0.28
18.24	3.94	0.19	1.10
17.30	4.66	0.27	0.59
10.75	4.16	0.36	0.45
12.56	4.01	0.39	0.07
18.83	3.18	0.45	0.18
20.30	2.99	0.09	0.07
18.47	4.98	0.10	0.19
18.29	4.27	0.34	0.62

avg_sed_hrs	avg_light_hrs	avg_moderate_hrs	avg_very_active_hrs
16.54	3.09	0.22	0.33

- Because the majority of this information is best looked at in hourly groups, this analysis is continued in Tableau where patterns can more easily be detected.